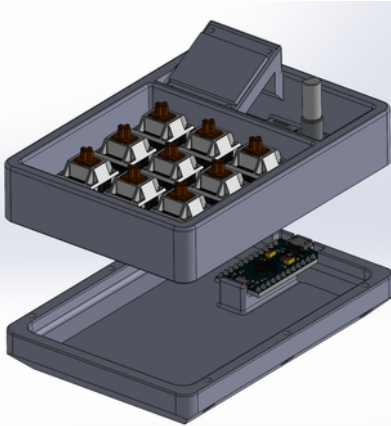


3x3 Custom Keyboard - Personal Project



What?

- Designed a custom keyboard for optimizing my workflow in **SolidWorks**.
- The objective was to **increase efficiency** while maintaining an intuitive and visually appealing design.



How?

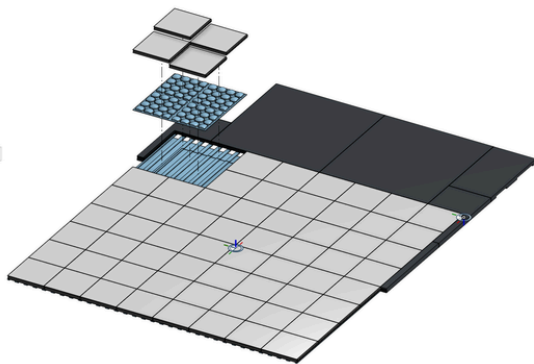
- Designed a 3D-printed enclosure with heat-set inserts to mount a **microcontroller**, **rotary encoder**, and **OLED screen**.
- Designed and ordered a **custom PCB** for the switches using **KiCad**.
- Developed **custom firmware** for communication between components.



Outcomes

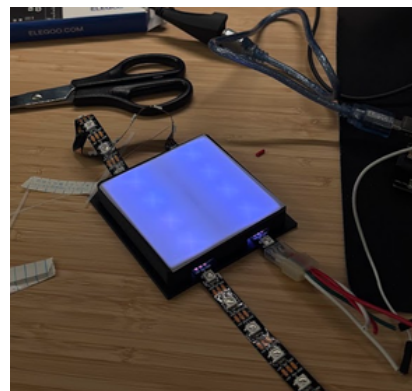
- Created a custom macropad with **39+ programmable functions**.
- Gained **end-to-end product design** experience, from CAD modelling to **electronics integration** and **firmware development**.

Custom LED-Embedded Chessboard - Spark!



What?

- Designed an LED-embedded chessboard for my design team's Smart Chess project.
- The objective was to **optimize the board thickness**, ensuring **structural strength** while maintaining a **thin profile for reliable magnetic attraction** on opposite sides of the board.
- Light from LEDs also had to diffuse minimally between board pieces.



How?

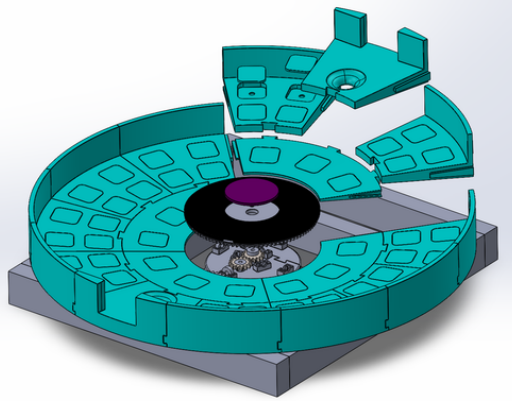
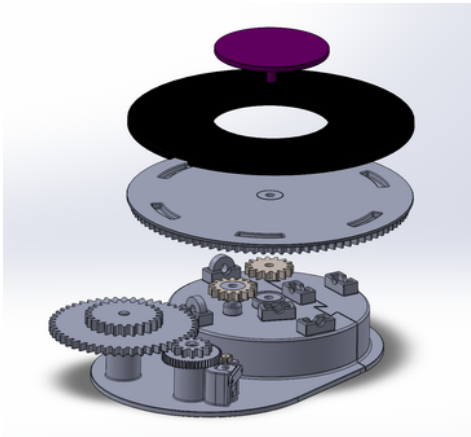
- Designed a system with three layers: a topper that connects to neighbouring toppers and prevented light bleed to other positions, a diffusion layer and a layer for holding LED strips.
- Used **Onshape** to **prototype** and test **10+ iterations** during the design process.
- Tested light diffusion using the FastLED Arduino library.



Outcomes

- Designed and manufactured an LED-embedded chessboard with a thickness of **7.7mm** and a deflection of less than **2mm**.
- Chessboard contains **1024+ LEDs** with little to no light bleeding between squares.

Hungry Hippos Arena - Spark!



What?

- Built a modular arena whose spinning centre scatters dropped balls.
- Goals: **maximum modularity and ball chaos**, while keeping a **600 mm outer diameter**.

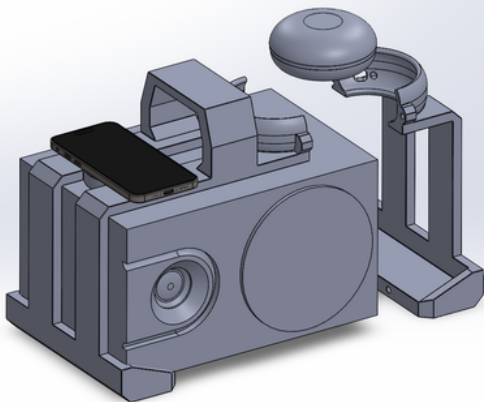
How?

- **Led 2 students through 3 spinner prototypes** to find the most chaotic design.
- Designed **counter-rotating dual spinners** driven by a single motor
- Designed a **1:130 gear reduction** to give one motor enough torque for both spinners.

Outcomes

- Delivered a fully modular 600 mm arena from **85+ 3D-printed parts**.
- Reached consistent spinner speeds of **70+ RPM**.
- [Watch the demo.](#)

Smart Bookshelf Speaker - Personal Project



What?

- Designed, and assembled a **smart speaker with wireless charging** from a broken Google Home Mini.
- Goals: **low cost**, simple aesthetics, and **better sound** than the original Mini.

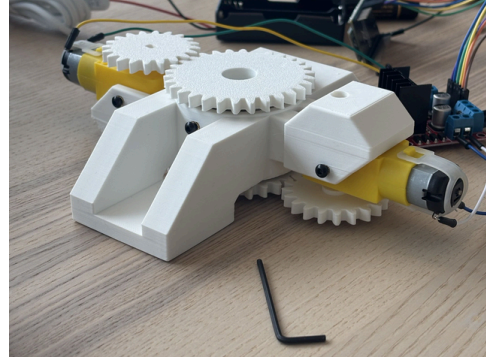
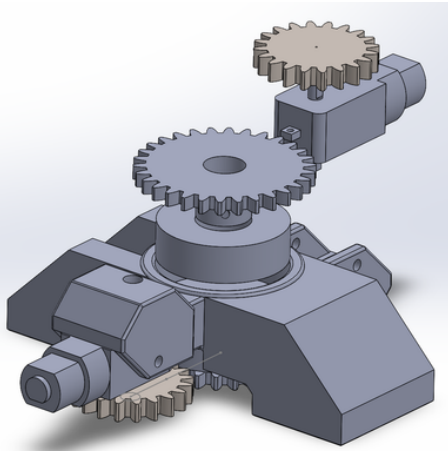
How?

- Thrifted a used 3-way bookshelf speaker for **\$15**.
- Salvaged a dead 2018 Home Mini and wired its amp output to the speaker terminals.
- **Modelled an enclosure** that holds the Mini and a MagSafe charger, with internal cable routing.

Outcomes

- Delivered a **smart speaker** with wireless charging for **under \$30**.
- Clear **improvement in volume and bass** over the stock Mini speaker.
- Hid rewiring inside the enclosure for a clean shelf presence.

Crater Crawl Player Prototype - Spark!



What?

- Built a prototype to showcase character movement for Spark's Crater Crawl project.
- Required 2 concentric rings controlled independently, with a hollow centre to hold the characters.
- Had to be **finished within a week** for design review and **reliable enough for live demos**.

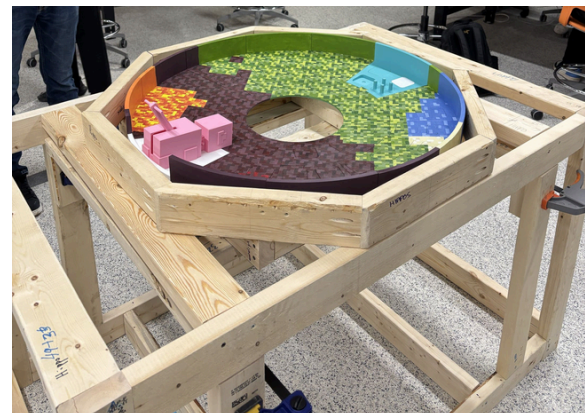
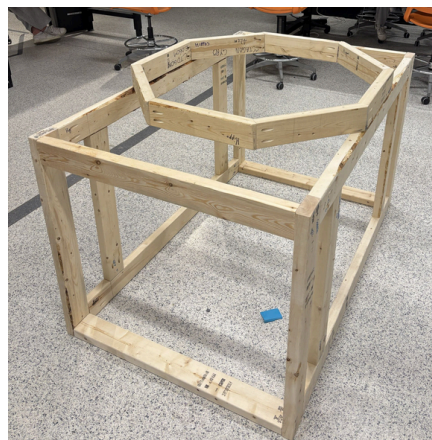
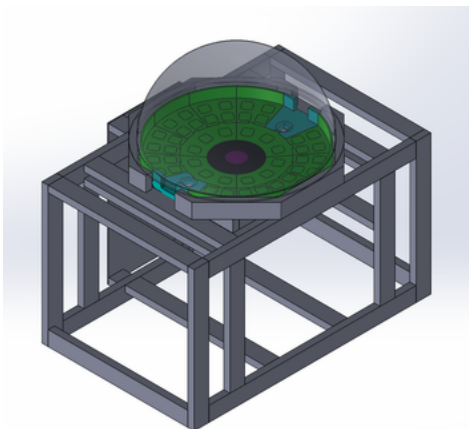
How?

- **Rapidly iterated on a 3D-printed enclosure** housing 2 DC motors and 2 large bearings.
- **Designed heat-set insert slots** so set screws could hold the gears and bearings in place.
- **Programmed ESP32 firmware** to drive both motors from a **Bluetooth controller**.

Outcomes

- Delivered a working prototype in **3 days**, ahead of the one-week design review deadline.
- Achieved **independent control of each ring** through the Bluetooth controller's buttons.
- Demoted it at the annual general meeting with **100+ participants** to recruit new members.

Hungry Hippos Frame - Spark!



What?

- Helped design and build a **custom 56" x 36" wooden frame** for our Hungry Hippos display
- Goals: **structural strength** and space to mount all electronics and mechanisms (arena, hippos, ball circulation)

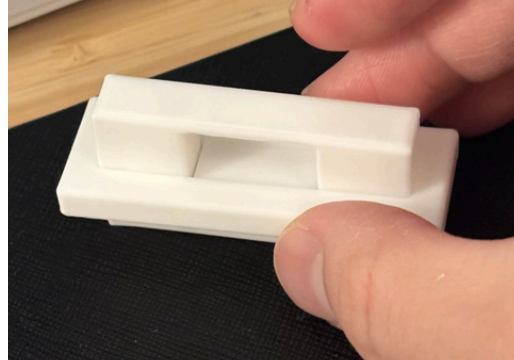
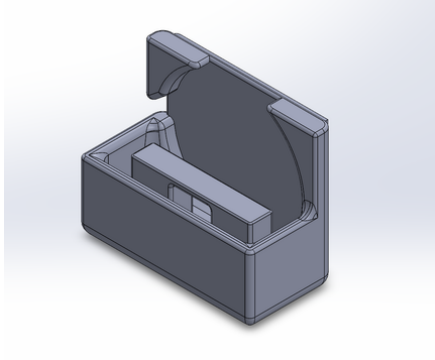
How?

- Collaborated with 3 sub-teams to merge our CAD and size the frame
- **Built a wooden octagon** to house the 3D-printed arena using **pocket holes and a miter saw**

Outcomes

- Delivered a frame from **40+ wooden parts** that stays stable under heavy force on the surface
- Gained hands-on experience with pocket-hole joints, miter saws, drills, impact drivers and jigsaws

AirPods Dock/Charger - Personal Project



What?

- Designed and modelled a custom dock and charger for my AirPods Pro to encourage myself to charge my AirPods more.
- The objective was to **maximize access speed, intuitiveness and simplicity**.

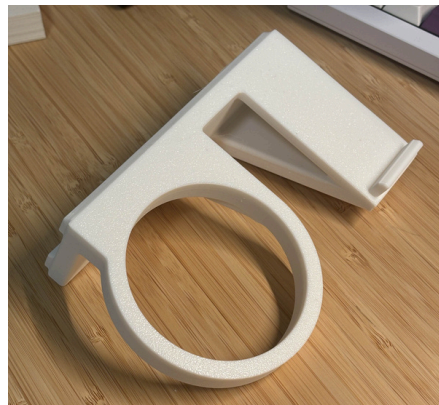
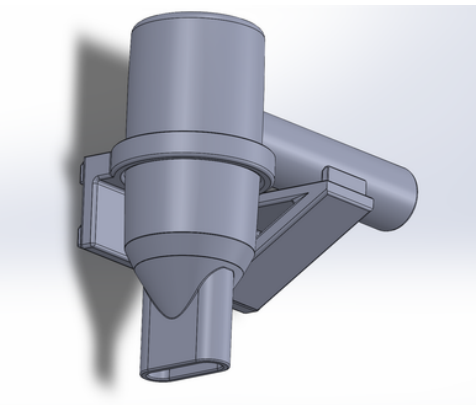
How?

- Researched and developed a **print-in-place mechanism** so that the entire part can be printed without assembly, despite containing 2 separate parts.
- Designed and modelled 2 prototypes to **ensure a perfect fit** for MagSafe charger and AirPods.

Outcomes

- My AirPods have always been charged before leaving the house for the past **6 months**.
- All relevant files and print instructions are **open-source** on **Thingiverse**.
- [Link to project](#).

Hair Dryer Holder - Personal Project



What?

- Designed a hair dryer holder so that I can hide my hair dryer within my bathroom closet.
- The objective was to design something with **sufficient support** while remaining **simple** and **fast to print**.

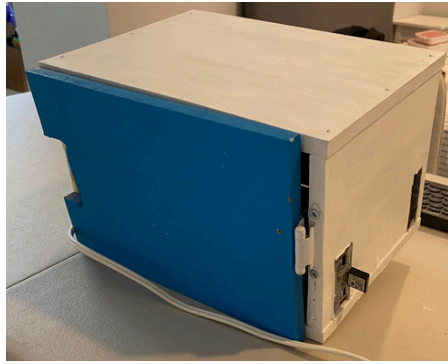
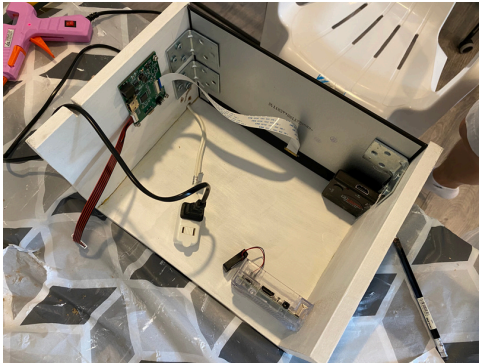
How?

- Measured and modelled the hair dryer to test fitting within a **SolidWorks assembly**.
- Strategically designed model to ensure **no supports** were needed, thus **saving filament** and keeping **print times low**.

Outcomes

- Space beside the bathroom sink is now **50% less cluttered**.
- Project received positive feedback, with **47 likes and 8 saves** on **Thingiverse**.
- [Link to project](#).

Smart Digital Photo Frame/Second Monitor - Personal Project



What?

- Designed a **smartphone-controllable** digital photo frame that can double as an external monitor.
- The objective was to **maximize utility** while keeping the **design simple** and unobtrusive on the desk.

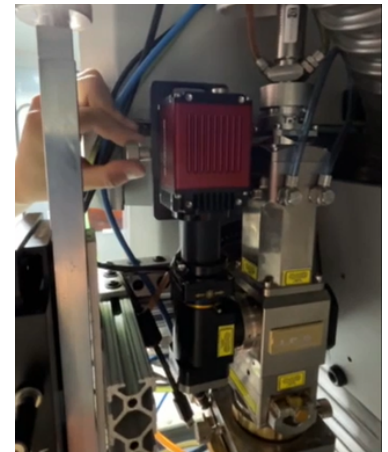
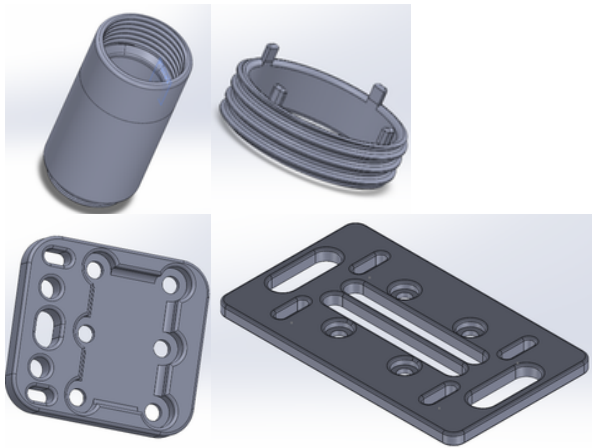
How?

- Researched and learned how to **woodwork** with **hand drills** and **saws** to create the enclosure.
- Learned **Python** to adapt and **debug** scripts from **GitHub**.

Outcomes

- Built a Digital photo frame with 32GB of expandable storage using USB ports.
- Created an enclosure that integrated a **Raspberry Pi**, cooling fan, 11" monitor, HDMI switcher, and monitor control board.
- Photo frame features can be controlled from any phone on the local network.

IR Camera Holder - LEMAM



What?

- Designed an adjustable lens mount and fixture for a \$27,000 IR camera.
- The objective was to create a **stable** fixture that had **smooth motion** and **fast print times**.

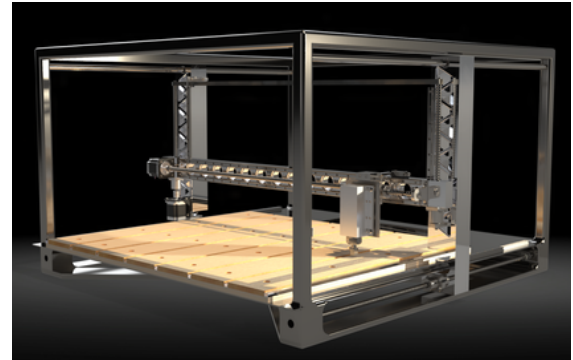
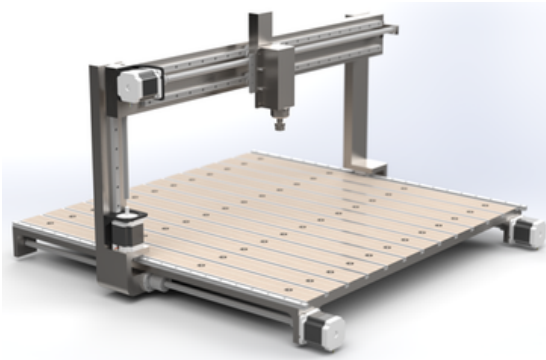
How?

- Designed a cap with **standardized** Sony threads to mount a lens filter onto the IR Camera.
- Measured mounting hole dimensions on the DED machine for design.
- Created a fixture pairing the IR camera with a linear guide mounted to the DED machine.

Outcomes

- IR camera now has **50+mm** of travel along the Z-axis for focusing.
- IR camera is stable, with less than **2mm** of wobble when secured.
- Net print time of under **2 hours** on the Prusa i3 mk3s+.

Custom CNC Machine - University of Toronto



What?

- Designed and modelled a large-format three-axis CNC router with a combination of **custom-designed parts** and **McMaster-Carr components**.
- The objective was to **maximize utility, modularity, and repairability** while **keeping costs low**.

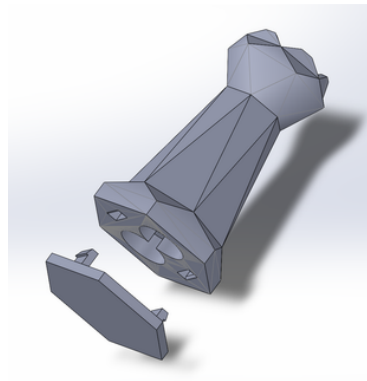
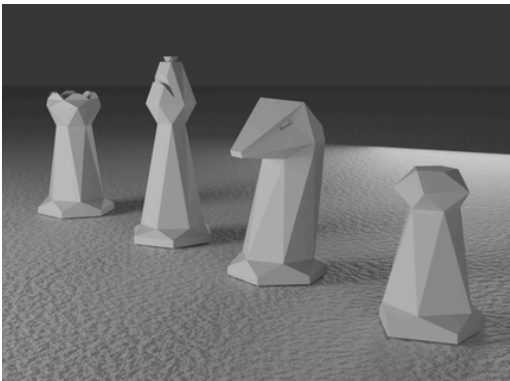
How?

- Researched components to **optimize for cost-efficiency** and performance based on **design specifications**.
- Developed and delivered a presentation to communicate **key design features and trade-offs**, effectively securing approval during the **engineering design briefing**.

Outcomes

- Produced a full CAD model with **229** parts and **400+** mates in **SolidWorks**.
- Created a comprehensive **design report**, including **engineering specifications**, component justification and a **bill of materials** outlining all components required to assemble the router.

Custom Chess Pieces with Magnets - Spark!



What?

- Designed 4 unique chess pieces with **internal magnets** for my team's Smart Chess project.
- The objective was to design something with a **low-poly aesthetic** that could house internal magnets.

How?

- Designed and modelled the chess pieces in **Blender**.
- Created renders of the design in **Blender** for marketing purposes.
- Created a **Blender to SolidWorks workflow** to utilize both organic modelling and technical refinement across both software tools.
- Designed a **3D printable clip-on cover** and magnet slots within **SolidWorks**.

Outcomes

- Printed **28 custom chess pieces**, each with 4 embedded magnets that let them be moved from below the chessboard.
- Pieces move smoothly across the board while keeping a clean low-poly aesthetic.